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Potassium hydroxide (KOH) solution is hung into the conical flask with the help of a thread and tightly close the one holed cork (Figure 14.11). Take a bent glass tube, the shorter end of which is inserted into the conical flask through the hole in the cork, while the longer end is dipped in a beaker containing water. Observe the position of initial water level in bent glass tube. This experimental setup is kept for two hours and the seeds were allowed to germinate. After two hours, the level of water rises in the glass tube. It is because, the $CO_2$ evolved during aerobic respiration by germinating seeds will be absorbed by KOH solution and the level of water will rise in the glass tube.

$$CO_2 + 2KOH \longrightarrow K_2CO_3 + H_2O$$

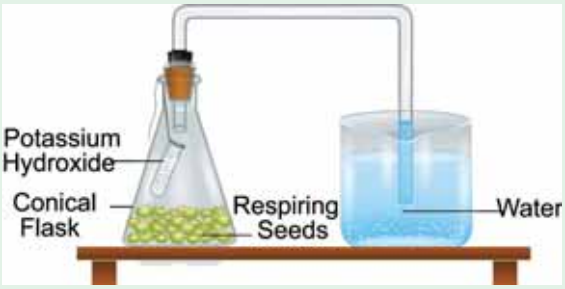



**Figure 14.11:** Demonstration of production of $CO_2$ during respiration

In the case of groundnut or bean seeds, the rise of water is relatively lesser because these seeds use fat and proteins as respiratory substrate and release a very small amount of $CO_2$. But in the case of wheat grains, the rise in water level is greater because they use carbohydrate as respiratory substrate. When carbohydrates are used as substrate, equal amounts of $CO_2$ and $O_2$ are evolved and consumed.

**Activity**

Take a test tube with some germinated seeds and fill with water. Keep this test tube after some time until liberation of $CO_2$. When the carbon dioxide from respiration is mixed to water, carbonic acid ($H_2CO_3$) is produced. Therefore, as more carbon dioxide is released, the solution becomes more acidic. You will see changes in pH as an indicator using blue litmus paper changed into red that respiration has occurred

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

## 14.7 Anaerobic Respiration

### 14.7.1 Fermentation

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Some organisms can respire in the absence of oxygen. This process is called **fermentation or anaerobic respiration** (Figure 14.12). There are three types of fermentation:

1. Alcoholic fermentation
2. Lactic acid fermentation
3. Mixed acid fermentation

#### 1. Alcoholic fermentation

The cells of roots in water logged soil respire by alcoholic fermentation because of lack of oxygen by converting pyruvic acid into ethyl alcohol and $CO_2$. Many species of yeast (*Saccharomyces*) also respire anaerobically. This process takes place in two steps:

(i) $2CH_3COCOOH \xrightarrow[TPP]{\text{Pyruvate decarboxylase}} 2CH_3CHO + 2CO_2\uparrow$
Pyruvic acid → Acetaldehyde

(ii) $2CH_3CHO + 2NADH + 2H^+ \xrightarrow{\text{Alcohol dehydrogenase}} 2CH_3CH_2OH + 2NAD^+$
Acetaldehyde → Ethyl alcohol